

Estimates of the In-Sample Error

- We obtained the following relation:

$$\mathbb{E}(\text{Err}_{in}) = \mathbb{E}(\overline{\text{Err}}) + \omega$$

- An estimate would then be:

$$\widehat{\text{Err}}_{in} = \overline{\text{Err}} + \hat{\omega}, \quad \text{where } \hat{\omega} \text{ is an estimate of the average optimism.}$$

- Under a squared loss, we obtained that:

$$\mathbb{E}(\text{Err}_{in}) = \mathbb{E}(\overline{\text{Err}}) + 2 \frac{d}{N} \sigma^2, \quad \text{where } d \text{ is the number of inputs.}$$

- A natural estimate is then:

$$\widehat{\text{Err}}_{in} = \overline{\text{Err}} + 2 \frac{d}{N} \hat{\sigma}^2, \quad \text{where } \hat{\sigma}^2 \text{ is an estimate of the noise variance.}$$

- Given a set of models indexed by a tuning parameter $f_{\alpha}(\mathbf{x})$, we have, for instance, when the log-likelihood loss function is used:

$$AIC(\alpha) = \overline{\text{Err}}(\alpha) + \frac{2}{N} d(\alpha) \hat{\sigma}^2$$

This provides an estimate of the test error curve, and we find the tuning parameter minimizing that criterion.